

aritate = nr. de argumente

$$2) U = (\exists x)P(x) \wedge (\exists x)Q(x) \rightarrow (\exists x)(P(x) \wedge Q(x))$$

$$I_2 = \langle D_2, m_2 \rangle$$

$$D_2 = \mathbb{R}$$

$$m_2(P): \mathbb{R} \rightarrow \{T, F\}, m_2(P)(x) = "x : 2"$$

$$m_2(Q): \mathbb{R} \rightarrow \{T, F\}, m_2(Q)(x) = "x > 0"$$

$$\begin{aligned} \bigvee^{I_2}(U) &= \bigvee^{I_2}((\exists x)P(x) \wedge (\exists x)Q(x) \rightarrow (\exists x)(P(x) \wedge Q(x))) = \\ &= \bigvee^{I_2}((\exists x)P(x) \wedge (\exists x)Q(x)) \rightarrow \bigvee^{I_2}((\exists x)P(x) \wedge Q(x)) = \\ &= \bigvee^{I_2}((\exists x)P(x)) \wedge \bigvee^{I_2}((\exists x)Q(x)) \rightarrow \bigvee^{I_2}((\exists x)P(x)) \wedge \bigvee^{I_2}(Q(x)) = \\ &= \end{aligned}$$

9.2.6. toate formule prenexe, Skolem, clauzate

$$(\exists x)(\neg(\exists y)P(y) \rightarrow (\forall y)(Q(y) \rightarrow R(x)))$$

$$P_1: \text{Înlocuim } \rightarrow, \leftrightarrow: A \rightarrow B \equiv \neg A \vee B$$

$$(\exists x)((\neg(\exists y)P(y) \vee (\forall y)(\neg Q(y) \vee R(x)))$$

P_2 : De Morgan (nu e cazul)

$$P_3: \text{Redenumim variabilele legate: } (\exists x)((\exists z)P(z) \vee (\forall y)(\neg Q(y) \vee R(x)))$$

P_4 : Extragera cuantificatorilor (2 variante):

$$U^{P_1} = (\exists x)(\exists z)(\forall y)(P(z) \vee \neg Q(y) \vee R(x))$$

$$U^{P_2} = (\exists x)(\forall y)(\exists z)(P(z) \vee \neg Q(y) \vee R(x))$$

P_5 : Eliminăm cuantificatorii existențiali:

$$x \leftarrow a, z \leftarrow b: U^{S_1} = (\forall y)(P(b) \vee \neg Q(y) \vee R(a))$$

$$x \leftarrow c, z \leftarrow f(y): U^{S_2} = (\forall y)(P(f(y)) \vee \neg Q(y) \vee R(c))$$

P_6 : Eliminăm cuantificatorii universali:

$$U^{SQ_1} = P(b) \vee \neg Q(y) \vee R(a) = U^{C_1}$$

$$U^{SQ_2} = P(f(y)) \vee \neg Q(y) \vee R(c) = U^{C_2}$$

P_7 : aducem la FNC (nu e cazul)

9.2. Aduceți la o formă prenexă, Skolem, clauzată:

$$(\forall x)(\exists y)((\exists x)P(z) \wedge (\forall u)(Q(x, u) \rightarrow (\exists z)Q(y, z)))$$

P_1 : înlocuim $\rightarrow, \leftrightarrow$: $A \rightarrow B \equiv \neg A \vee B$

$$(\forall x)(\exists y)((\exists x)P(z) \wedge (\forall u)(\neg Q(x, u) \vee (\exists z)Q(y, z)))$$

P_2 : De Morgan (nu e cazul)

P_3 : Redenumim variabilele legate:

$$(\forall x)(\exists y)((\exists w)P(x) \wedge (\forall u)(\neg Q(x, u) \vee (\exists z)Q(y, z)))$$

